



School of Computer Science

COMP47470: Big Data Programming

**Neo4j NOSQL DBMS and Cypher Query
Language**

Lab Practical 1 Solutions

Module Coordinator	Ravi Reddy Manumachu
Email	ravi.manumachu@ucd.ie

Neo4j Movies

1.

```
MATCH (m:Movie)<-[:ACTED_IN]-(a:Person)
WITH m, count(DISTINCT a) AS castSize
WHERE castSize = 2
RETURN m.title AS movie, castSize
ORDER BY movie;
```

2.

```
MATCH (a:Person)-[:ACTED_IN]->(m:Movie)
WHERE m.released IS NOT NULL
WITH a, m.released AS year, count(m) AS moviesInYear
WHERE moviesInYear > 1
RETURN a.name AS actor, year, moviesInYear
ORDER BY moviesInYear DESC, actor;
```

3.

```
MATCH (m:Movie)
WHERE NOT EXISTS {
  MATCH (a:Person)-[:ACTED_IN]->(m)
}
RETURN m.title AS movie
ORDER BY movie;
```

Neo4j Stations and Stops

1.

```
MATCH p = SHORTEST 1
(:Station {name: "Brixton"})
()-[:LINK]-(:Station))+
(:Station {name: "Clapham Junction"})
RETURN [station IN nodes(p) | station.name] AS route,
length(p) AS hops;
```

2.

```
MATCH p = SHORTEST 1
(:Station {name: "Brixton"})
()-[:LINK]-(:Station))+
(:Station {name: "Clapham Junction"})
RETURN [station IN nodes(p) | station.name] AS route,
reduce(total = 0.0, r IN relationships(p) | total + r.distance) AS totalDistanceKm
ORDER BY totalDistanceKm;
```